7.0 Introduction

Many sub systems are assembled to form an automobile. With this automobile, the passenger and goods are transported from one location to another location. During travelling, if the road shocks are transmitted to the chassis without absorption, then it will provide uncomfortable ride to the passengers as well as lead to failure of vehicle parts and goods transported. To prevent this, suspension systems are used and it is located between the axle housing and the vehicle chassis. Thus, the suspension system in automobile is a very important in deciding vehicle drive comfort and the stability of the vehicle.

7.0.1 Functions of suspension system

1. To reduce the effect of the shock and provide comfortable ride.
2. To prevent the damage of the vehicle parts when the vehicle travel over a rough road.
3. To prevent the steering system from road shocks
4. To prevent the occurrence of roll over when the vehicle travel over an irregular road.
5. To connect the chassis and frame.

7.0.2 Properties of Suspension system

1. The initial cost and the maintenance cost should be low
2. Wear and tear of the parts should be low
3. Weight should be less

7.1 Spring

Spring plays a vital role in a suspension system. Spring is located between the wheel axle and the vehicle body or frame. When the vehicle runs over a rough road, the spring absorb the road shock by compression and expansion. Some part of the shocks are absorbed by the tyres.

7.1 Spring Types

7.1.1 Steel Springs
7.1.2 Rubber Spring
7.1.3 Air Spring
7.1.4 Plastic Spring

7.1.1 Steel Springs types

1. Leaf spring
2. Coil Spring
3. Stabilizer Bar
4. Torsion Bar

7.1.1 Leaf Spring

Introduction

This type of springs is widely used in light and heavy commercial vehicles.

Construction

A leaf spring is built with number of leaves. The leaves are placed one over the other, and are held together by clamps and a bolt at the center. The leaves are usually given an initial curvature or chambered, so that they will tend to straighten under load. The spring is clamped to the housing by means of U – bolt. The longest leaf known as the main leaf or master leaf has its ends formed in the shape of an eye.

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